

Beyond univariate GP

Sparse MLP, U-Net and real multivariate targets

Egor Serov · Kirill Frolov · Daniil Koblov · Vasilii Lyamin

This extension adds a fixed-mask sparse MLP, a temporal U-Net, genuinely joint multivariate forecasting, and two real datasets. Dense MLP and S4 are retrained as controls. The original univariate synthetic experiments remain unchanged.

Completed comparison	Protocol
96 MPS fits	4 datasets × 4 heads × 2 sources × 3 seeds
Capacity and training	Approximately 64k active parameters; 2,000 updates; best validation checkpoint
Conditional forecasts	64 test contexts × 32 generated futures; 64 Euler steps
External validity	ETTh1 from Sundial evaluation; Exchange from TSFlow evaluation
Verification	Fresh-data reconstruction, independent metric recomputation, CPU/MPS model checks

Lowest observed mean CRPS in each dataset

Dataset	Neural cell	CRPS ± seed SD	Best baseline	CRPS
Correlated GP: three channels	U-Net / white	0.3215 ± 0.0058	ridge residual bootstrap	0.1929
ETTh1: oil temperature	Sparse MLP / white	0.1409 ± 0.021	ridge residual bootstrap	0.09333
ETTh1: seven channels	Sparse MLP / white	0.3585 ± 0.017	ridge residual bootstrap	0.287
Exchange: eight channels	Sparse MLP / white	0.2837 ± 0.023	ridge residual bootstrap	0.06778

These are descriptive minima among tested cells, not statistically established winners. CRPS is in train-standardized units and should only be compared within a dataset. Baselines are fitted once and have no optimization-seed SD. White means temporally white with the same cross-channel correlation as the GP source.

Keep the baseline comparison in view. On 4 of the four datasets, a simple baseline has lower CRPS than the best mean neural cell. Architecture ranking alone therefore does not establish forecasting usefulness. All scores and the uncertainty/calibration checks are reported below.

A source-dependent crossover appears on ETTh1 OT. S4 has lower CRPS than dense MLP with the GP source (0.144 vs 0.195), but higher CRPS with the white source (0.170 vs 0.142). Its GP-source residual velocity MSE is nevertheless worse. This is evidence about forecasting at this budget, not confirmation of the original small-PC failure mechanism.

This is a fixed-budget generalization study, not a full reproduction of pretrained Sundial or TSFlow. Real-data and multivariate results do not replace the original assignment's univariate rank-two geometry or its declared success rule.

What changes, and what stays comparable

Head	Architecture	Sparse/temporal detail
Dense MLP	Joint flattened horizon and channels; adaptive LayerNorm	Unstructured learned affine maps
Sparse MLP	Same family, fixed approximately 50% masks on all linear maps	Wider to match active capacity; dense kernels and about twice allocated weights
S4	Three rank-one HiPPO/NPLR temporal blocks with channel mixing	Reuses the independently validated project S4 operator
U-Net	Three temporal resolutions, skip connections and conditional convolutions	Two down/up stages; no convolution over flattened channel indices

Dataset	L → F	Train / val / test windows	Active counts: MLP / sparse / S4 / U-Net
Correlated GP: three channels	32 → 16	8192/1024/1024	63,913 / 63,395 / 64,212 / 62,403
ETTh1: oil temperature	48 → 24	2143/118/118	62,999 / 64,105 / 63,966 / 61,621
ETTh1: seven channels	48 → 24	2143/118/118	64,883 / 64,386 / 64,519 / 66,127
Exchange: eight channels	32 → 16	5264/45/92	64,482 / 62,923 / 64,248 / 64,958

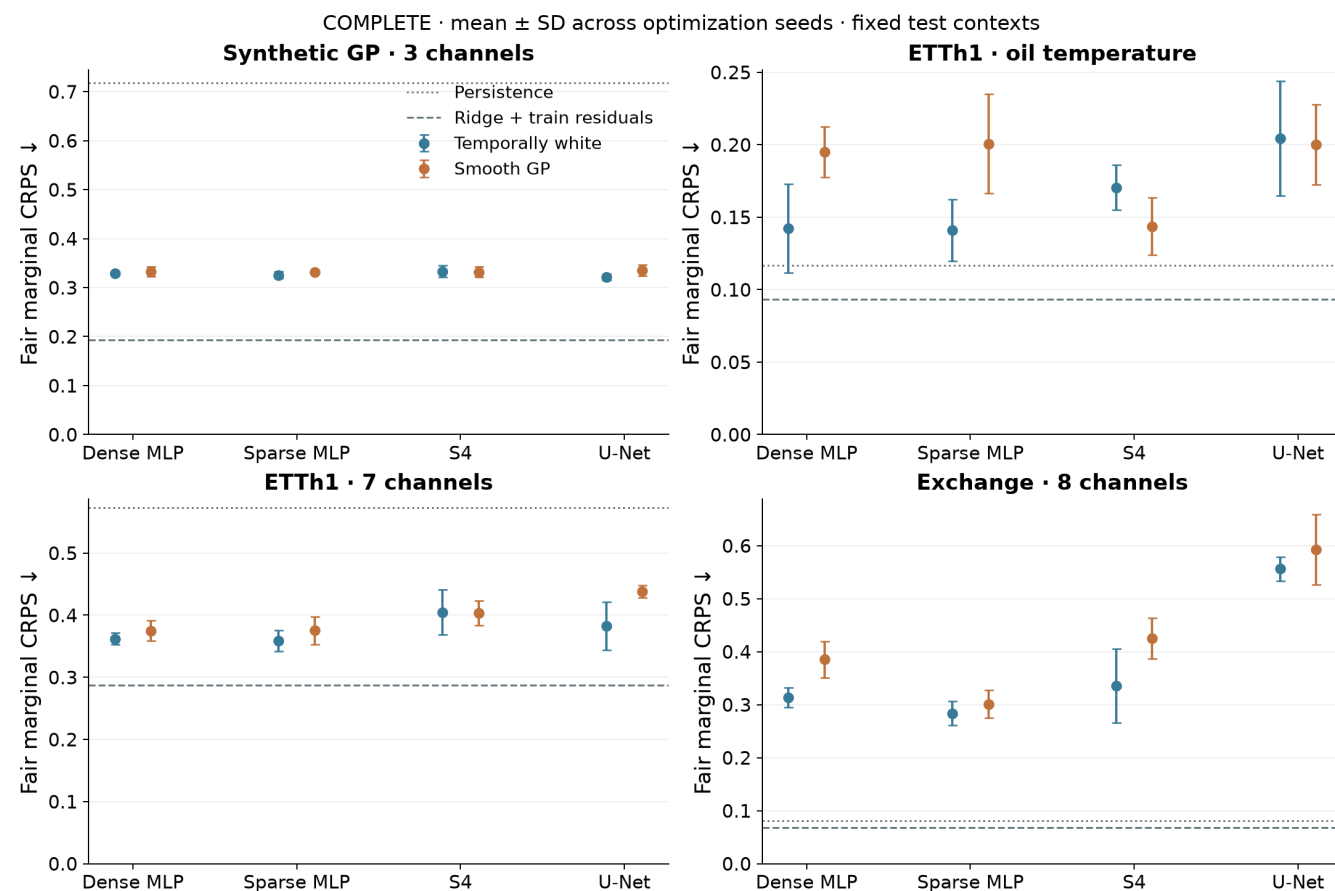
Information and scaling. Every head receives the same full history through an identity encoder. All channels use fixed training-only channel mean and standard deviation; no validation or test value enters fitting. Training targets may enter training statistics and PCA. Unlike the original experiment, this extension does not apply per-window history normalization.

Sources. Independent source/target pairs use the same train-estimated channel correlation. The white source has identity temporal covariance; the GP source has an RBF temporal kernel of lengthscale 8. Both have unit marginal source variance. Cross-channel correlation is held fixed while temporal covariance changes.

Optimization. Raw velocity MSE, AdamW learning rate 0.001, batch 128, gradient clipping 1, 2,000 updates. Validation is checked every 250 updates. There is no head-specific rate search here. Equal active capacity and updates do not imply equal computational cost or equal convergence.

ETTh1 uses the 12/4/4-month row split and 48→24 forecasting; Exchange uses a 70/10/20% chronological split and 32→16 forecasting. Whole history+future windows lie within their split. Test targets do not overlap, but temporal dependence remains. Data revisions, SHA-256 checksums, exact starts and scalars are saved.

Compare predictions on the same targets



Marginal fair CRPS. Lower is better. Markers/error bars summarize three optimization seeds on the same 64 held-out contexts. Compare sources within a dataset, not across datasets.

Mean CRPS difference: GP source minus white source

Head	GP data (3C)	ETTh1 (1C)	ETTh1 (7C)	Exchange (8C)
Dense MLP	+0.0040	+0.0528	+0.0129	+0.0715
Sparse MLP	+0.0060	+0.0597	+0.0166	+0.0174
S4	-0.0012	-0.0267	-0.0011	+0.0901
U-Net	+0.0138	-0.0044	+0.0557	+0.0365

Negative values favor the GP source within that head and dataset; these are descriptive mean differences. Joint energy evaluates the whole horizon $\times$ channel vector and is divided by $\sqrt{F \times C}$. It is reported alongside CRPS in every dataset table and as a separate vector figure.

Ensemble scores use off-diagonal corrections for 32 samples. The full CSV includes channel-sum CRPS, point MSE, 90% interval coverage/width and roughness. A visually smooth path can still be an inaccurate or poorly calibrated forecast.

Correlated GP: three channels

Head	Source	CRPS ↓	Joint energy ↓	Mean MSE ↓	90% coverage
Dense MLP	white	0.3288 ± 0.0037	0.4301 ± 0.005	0.4556 ± 0.0068	84.8%
Sparse MLP	white	0.3257 ± 0.0072	0.4272 ± 0.0076	0.4499 ± 0.014	85.4%
S4	white	0.333 ± 0.012	0.4357 ± 0.014	0.4548 ± 0.029	81.4%
U-Net	white	0.3215 ± 0.0058	0.4216 ± 0.0069	0.4321 ± 0.014	80.2%
Dense MLP	gp	0.3328 ± 0.0096	0.4309 ± 0.0099	0.4593 ± 0.019	82.9%
Sparse MLP	gp	0.3318 ± 0.0021	0.4312 ± 0.0038	0.4579 ± 0.0097	82.6%
S4	gp	0.3318 ± 0.011	0.4297 ± 0.01	0.4569 ± 0.017	81.7%
U-Net	gp	0.3353 ± 0.012	0.4336 ± 0.013	0.465 ± 0.023	82.6%

Mean $\pm$ sample SD over three training seeds. Coverage is the marginal proportion of target coordinates inside central 90% sample intervals. It is not simultaneous whole-path coverage.

Connection to the original stress-test metrics

Head	Source	PC1 velocity MSE	Residual velocity MSE	Roughness ratio
Dense MLP	white	6.353 ± 0.054	0.2022 ± 0.0043	1.12 ± 0.017
Sparse MLP	white	6.278 ± 0.077	0.2137 ± 0.0038	1.181 ± 0.033
S4	white	6.002 ± 0.23	0.2001 ± 0.0025	1.037 ± 0.031
U-Net	white	5.516 ± 0.061	0.1696 ± 0.004	0.9535 ± 0.004
Dense MLP	gp	15.95 ± 0.5	0.2892 ± 0.0044	0.9809 ± 0.019
Sparse MLP	gp	15.9 ± 0.26	0.2946 ± 0.0033	1.203 ± 0.023
S4	gp	16.17 ± 0.74	0.3148 ± 0.0085	0.9492 ± 0.05
U-Net	gp	15.47 ± 0.45	0.2968 ± 0.011	1.002 ± 0.051

PCA is fitted to training velocities separately for each source. Residual error is the per-component average over PCs 3 onward in joint $F \times C$ coordinates. Roughness is computed along time only, with target-normalized ratio 1 meaning equal average absolute adjacent increments. Velocity labels differ across sources; lower cross-source velocity MSE alone does not establish a better forecast.

Baseline	CRPS	MSE	90% coverage
persistence	0.7182	0.9753	0.0%
ridge	0.2834	0.2134	0.0%
ridge residual bootstrap	0.1929	0.2039	86.4%

Ridge alpha 0.1 was chosen on validation MSE. Residual bootstrap resamples entire training residual vectors and can understate uncertainty. Deterministic baselines have degenerate intervals.

ETTh1: oil temperature

Head	Source	CRPS ↓	Joint energy ↓	Mean MSE ↓	90% coverage
Dense MLP	white	0.1422 ± 0.03	0.1603 ± 0.032	0.06736 ± 0.028	85.0%
Sparse MLP	white	0.1409 ± 0.021	0.1587 ± 0.022	0.06522 ± 0.018	86.7%
S4	white	0.1704 ± 0.015	0.1904 ± 0.014	0.1009 ± 0.013	75.0%
U-Net	white	0.2045 ± 0.04	0.2251 ± 0.043	0.136 ± 0.054	68.3%
Dense MLP	gp	0.195 ± 0.017	0.2156 ± 0.015	0.1261 ± 0.012	69.1%
Sparse MLP	gp	0.2006 ± 0.034	0.2266 ± 0.038	0.1442 ± 0.029	69.0%
S4	gp	0.1437 ± 0.02	0.1639 ± 0.021	0.07274 ± 0.02	79.9%
U-Net	gp	0.2002 ± 0.028	0.2188 ± 0.029	0.1149 ± 0.026	64.6%

Mean ± sample SD over three training seeds. Coverage is the marginal proportion of target coordinates inside central 90% sample intervals. It is not simultaneous whole-path coverage.

Connection to the original stress-test metrics

Head	Source	PC1 velocity MSE	Residual velocity MSE	Roughness ratio
Dense MLP	white	1.598 ± 0.36	0.1175 ± 0.0051	2.059 ± 0.067
Sparse MLP	white	1.507 ± 0.29	0.1168 ± 0.0069	2.086 ± 0.07
S4	white	2.555 ± 0.39	0.1167 ± 0.0035	1.78 ± 0.29
U-Net	white	3.329 ± 0.79	0.1125 ± 0.0034	1.735 ± 0.051
Dense MLP	gp	10.29 ± 0.95	0.03569 ± 0.0036	0.8708 ± 0.029
Sparse MLP	gp	11.46 ± 4.6	0.0423 ± 0.0029	1.56 ± 0.2
S4	gp	6.123 ± 1.5	0.03983 ± 0.0025	0.6806 ± 0.063
U-Net	gp	11.06 ± 3.3	0.03947 ± 0.0022	0.996 ± 0.011

PCA is fitted to training velocities separately for each source. Residual error is the per-component average over PCs 3 onward in joint F×C coordinates. Roughness is computed along time only, with target-normalized ratio 1 meaning equal average absolute adjacent increments. Velocity labels differ across sources; lower cross-source velocity MSE alone does not establish a better forecast.

Baseline	CRPS	MSE	90% coverage
persistence	0.1169	0.02573	4.2%
ridge	0.1207	0.02535	0.0%
seasonal naive	0.1575	0.03989	2.3%
ridge residual bootstrap	0.09333	0.02782	96.8%

Ridge alpha 10 was chosen on validation MSE. Residual bootstrap resamples entire training residual vectors and can understate uncertainty. Deterministic baselines have degenerate intervals.

ETTh1: seven channels

Head	Source	CRPS ↓	Joint energy ↓	Mean MSE ↓	90% coverage
Dense MLP	white	0.3621 ± 0.0091	0.4696 ± 0.0094	0.4675 ± 0.02	92.4%
Sparse MLP	white	0.3585 ± 0.017	0.4671 ± 0.022	0.4688 ± 0.047	90.5%
S4	white	0.4045 ± 0.036	0.5483 ± 0.035	0.5982 ± 0.07	70.8%
U-Net	white	0.3825 ± 0.038	0.5403 ± 0.045	0.5734 ± 0.085	70.1%
Dense MLP	gp	0.375 ± 0.016	0.5069 ± 0.024	0.5112 ± 0.046	72.6%
Sparse MLP	gp	0.3751 ± 0.022	0.5105 ± 0.022	0.5115 ± 0.026	72.6%
S4	gp	0.4033 ± 0.02	0.542 ± 0.031	0.5954 ± 0.063	74.5%
U-Net	gp	0.4382 ± 0.0097	0.6186 ± 0.025	0.7245 ± 0.048	67.3%

Mean ± sample SD over three training seeds. Coverage is the marginal proportion of target coordinates inside central 90% sample intervals. It is not simultaneous whole-path coverage.

Connection to the original stress-test metrics

Head	Source	PC1 velocity MSE	Residual velocity MSE	Roughness ratio
Dense MLP	white	5.643 ± 0.67	1.036 ± 0.013	3.212 ± 0.044
Sparse MLP	white	5.901 ± 1.8	1.002 ± 0.017	2.869 ± 0.02
S4	white	12.02 ± 5.6	0.832 ± 0.037	1.471 ± 0.054
U-Net	white	6.314 ± 0.76	0.8316 ± 0.051	1.111 ± 0.069
Dense MLP	gp	26.06 ± 5.6	0.8011 ± 0.032	0.6249 ± 0.051
Sparse MLP	gp	21.53 ± 0.63	0.7916 ± 0.075	1.102 ± 0.083
S4	gp	34.86 ± 9.2	0.7579 ± 0.048	0.452 ± 0.025
U-Net	gp	23.7 ± 4.7	0.9026 ± 0.11	0.5263 ± 0.057

PCA is fitted to training velocities separately for each source. Residual error is the per-component average over PCs 3 onward in joint F×C coordinates. Roughness is computed along time only, with target-normalized ratio 1 meaning equal average absolute adjacent increments. Velocity labels differ across sources; lower cross-source velocity MSE alone does not establish a better forecast.

Baseline	CRPS	MSE	90% coverage
persistence	0.5727	0.9507	2.6%
ridge	0.3852	0.334	0.0%
seasonal naive	0.3868	0.4133	2.7%
ridge residual bootstrap	0.287	0.3395	82.8%

Ridge alpha 100 was chosen on validation MSE. Residual bootstrap resamples entire training residual vectors and can understate uncertainty. Deterministic baselines have degenerate intervals.

Exchange: eight channels

Head	Source	CRPS ↓	Joint energy ↓	Mean MSE ↓	90% coverage
Dense MLP	white	0.3137 ± 0.019	0.3819 ± 0.024	0.3059 ± 0.031	78.7%
Sparse MLP	white	0.2837 ± 0.023	0.3472 ± 0.026	0.2531 ± 0.035	83.4%
S4	white	0.3354 ± 0.07	0.3934 ± 0.084	0.2909 ± 0.11	47.6%
U-Net	white	0.5562 ± 0.023	0.668 ± 0.038	0.7415 ± 0.098	41.4%
Dense MLP	gp	0.3852 ± 0.035	0.4451 ± 0.033	0.3126 ± 0.046	33.1%
Sparse MLP	gp	0.3011 ± 0.026	0.364 ± 0.029	0.2441 ± 0.039	56.1%
S4	gp	0.4255 ± 0.038	0.5048 ± 0.052	0.4945 ± 0.098	59.3%
U-Net	gp	0.5927 ± 0.067	0.7105 ± 0.078	0.8434 ± 0.19	41.5%

Mean ± sample SD over three training seeds. Coverage is the marginal proportion of target coordinates inside central 90% sample intervals. It is not simultaneous whole-path coverage.

Connection to the original stress-test metrics

Head	Source	PC1 velocity MSE	Residual velocity MSE	Roughness ratio
Dense MLP	white	22.48 ± 8.8	0.5539 ± 0.049	19.92 ± 0.79
Sparse MLP	white	20.09 ± 14	0.5439 ± 0.072	20.21 ± 1.7
S4	white	27.25 ± 17	0.4728 ± 0.2	10.79 ± 2.8
U-Net	white	30.2 ± 13	0.916 ± 0.037	14.87 ± 2.4
Dense MLP	gp	32.81 ± 7.3	0.8828 ± 0.15	1.725 ± 0.028
Sparse MLP	gp	32.45 ± 13	0.6289 ± 0.049	8.816 ± 1
S4	gp	30.17 ± 8.9	0.7245 ± 0.07	1.741 ± 0.16
U-Net	gp	79.95 ± 45	1.115 ± 0.055	2.98 ± 0.3

PCA is fitted to training velocities separately for each source. Residual error is the per-component average over PCs 3 onward in joint F×C coordinates. Roughness is computed along time only, with target-normalized ratio 1 meaning equal average absolute adjacent increments. Velocity labels differ across sources; lower cross-source velocity MSE alone does not establish a better forecast.

Baseline	CRPS	MSE	90% coverage
persistence	0.08099	0.01457	1.1%
ridge	0.09399	0.01895	0.0%
ridge residual bootstrap	0.06778	0.01922	81.5%

Ridge alpha 0.1 was chosen on validation MSE. Residual bootstrap resamples entire training residual vectors and can understate uncertainty. Deterministic baselines have degenerate intervals.

Does observing other channels help OT?

Both ETTh1 experiments use the same held-out timestamps and OT training scaler. The joint model is rescored on its OT output alone. Comparing its seven-channel average against the OT-only model would answer a different question.

Training target	Head	Source	OT CRPS	OT mean MSE
ETTh1 OT-only	Dense MLP	white	0.1422 ± 0.03	0.06736 ± 0.028
ETTh1 joint	Dense MLP	white	0.2609 ± 0.017	0.1961 ± 0.035
ETTh1 OT-only	Dense MLP	gp	0.195 ± 0.017	0.1261 ± 0.012
ETTh1 joint	Dense MLP	gp	0.2813 ± 0.038	0.2224 ± 0.044
ETTh1 OT-only	Sparse MLP	white	0.1409 ± 0.021	0.06522 ± 0.018
ETTh1 joint	Sparse MLP	white	0.2817 ± 0.072	0.2434 ± 0.14
ETTh1 OT-only	Sparse MLP	gp	0.2006 ± 0.034	0.1442 ± 0.029
ETTh1 joint	Sparse MLP	gp	0.2602 ± 0.067	0.2058 ± 0.085
ETTh1 OT-only	S4	white	0.1704 ± 0.015	0.1009 ± 0.013
ETTh1 joint	S4	white	0.1971 ± 0.019	0.1171 ± 0.019
ETTh1 OT-only	S4	gp	0.1437 ± 0.02	0.07274 ± 0.02
ETTh1 joint	S4	gp	0.2251 ± 0.041	0.154 ± 0.053
ETTh1 OT-only	U-Net	white	0.2045 ± 0.04	0.136 ± 0.054
ETTh1 joint	U-Net	white	0.1963 ± 0.0089	0.1115 ± 0.01
ETTh1 OT-only	U-Net	gp	0.2002 ± 0.028	0.1149 ± 0.026
ETTh1 joint	U-Net	gp	0.3003 ± 0.053	0.2409 ± 0.062

A joint model predicts all seven channels with approximately the same total active-parameter budget as the univariate model. This is an operational comparison, not a clean causal estimate of adding information alone: output dimension, channel mixing and parameter allocation also change.

OT-only comparisons have paired seed differences in [output/extension-analysis/summary.json](#). The separate full-dataset head/source comparisons also have exploratory context intervals: circular moving blocks of four selected contexts, with block length eight as sensitivity for real data. These condition on the selected test segment and fixed trained seeds; neither three seeds nor two real datasets support a universal architecture claim.

What the checks establish

Validation variance share	Target E2	White velocity E2	GP velocity E2	GP velocity E2C
Correlated GP: three channels	0.710	0.396	0.678	0.969
ETTh1: oil temperature	0.958	0.265	0.908	0.908
ETTh1: seven channels	0.386	0.210	0.374	0.921
Exchange: eight channels	0.704	0.153	0.595	0.986

Training-fitted PCA. Multivariate top-2C and per-channel spectra supplement top-2 energy; channel modes can increase joint rank even when each channel is smooth. The original rank-two gate is not imposed on generalization experiments. Full spectra and plots are supplied separately.

Optimization limit. 25/96 runs select the last checked step. This alone neither proves nor disproves convergence; the full validation curves are supplied. Results describe a 2,000-update budget and one learning rate, not best achievable performance.

Euler 64 → 128 check	Max endpoint RMSE	Max CRPS change
Correlated GP: three channels	0.00611	0.000224
ETTh1: oil temperature	0.00401	0.000302
ETTh1: seven channels	0.00438	0.000301
Exchange: eight channels	0.00291	0.00137

The solver check covers every head/source at seed 31, first eight selected contexts. It is a numerical sensitivity check, not a guarantee for all contexts or larger step counts.

Independent checks. The audit reconstructs data and train-only transforms, verifies chronological splits, source covariance/PCA, best-checkpoint selection and file hashes, and independently recomputes all saved forecast and velocity metrics. Saved MPS checkpoint outputs are also reproduced on CPU. Full-model forward/backward/update parity passed for all four heads at 1, 3, 7 and 8 channels.

Reproduce with `scripts/fetch_extension_data.py`, `python -m fm_stress.extended_experiment`, `scripts/check_extension.py`, and `python -m fm_stress.analyze_extension`. Use the project virtual environment and MPS access. Exact commands and interpretation are in `docs/extension_protocol.md`; findings are in `docs/extension_findings.md`.

Primary sources: [Sundial / TimeFlow](#), [TSFlow](#), including multivariate Appendix B.8, [ETDataset](#), [Exchange data](#). This study uses their motivating ideas and datasets, with explicitly different small-model protocols.